

Part 1: Foundations

Block 1: What a data analyst does

- A data analyst looks at past data to explain **what happened and why**. (Data science, covered on Day 2, goes on to predict what will happen.)
- The workflow in five steps: **Ask a question, Collect data, Clean it, Analyse it, Visualise and communicate**.
- Typical tools: Excel, SQL, Python, and BI tools such as Power BI and Tableau.
- Real questions an analyst answers: Which region earns the least profit? Which products lose money? Are sales growing month by month?

Block 2: Data basics and cleaning

- Rows are records and columns are fields. Data types: numbers, text, dates. A wrong type (a date stored as text) breaks charts.
- Four common data problems: **missing values, duplicate rows, wrong data types, inconsistent spelling** (for example "UK", "U.K.", "United Kingdom").
- Why cleaning comes first: wrong data gives wrong dashboards ("garbage in, garbage out").
- Key terms to teach now (needed for both tools): **dimension** (a category such as Region), **measure** (a number you add up, such as Sales), **KPI** (a headline number such as total profit).
- Short demo: open the CSV in Excel, scroll through it and point out the columns, data types and any problems you can find.

Block 3: Power BI vs Tableau

Explain that both tools turn data into interactive dashboards, and that learning one makes the other much easier.

	Power BI	Tableau
Cost to learn	Power BI Desktop is free	Tableau Public is free (workbooks you save are public)
Runs on	Windows (Desktop app)	Windows and Mac
Data cleaning	Built-in Power Query editor	Basic cleaning in the data source tab; heavier cleaning is usually done before
Calculations	DAX formulas for measures	Calculated fields
Known for	Microsoft and Excel integration, business reporting	Fast visual exploration, flexible and polished charts

Do not upload confidential data to Tableau Public, because saved workbooks are visible to everyone.

Part 2: Power BI hands-on

Goal: by the end, every student has a one-page sales dashboard with KPI cards, a trend line, a category chart and slicers. Do the steps in order, and pause after each one to check that students are keeping up.

Step	What to do
Tour the interface	Open Power BI Desktop. Show the three views on the left: Report (build visuals), Data (see the table), Model (relationships). Show the Fields and Visualizations panes.
Load and clean data (Power Query)	Home > Get data > Text/CSV, choose the file, click Transform Data . Check that Order Date is a Date and Sales, Profit and Discount are numbers. Remove duplicate rows and rows with blank values, rename any unclear columns, then Close and Apply .
Write three measures (DAX)	Right-click the table > New measure. Create: Total Sales = SUM(Orders[Sales]) Total Profit = SUM(Orders[Profit]) Profit Margin % = DIVIDE([Total Profit], [Total Sales]) Replace "Orders" with the name of your table. Explain: a measure recalculates automatically when filters change.
Build the visuals	Three Card visuals (Total Sales, Total Profit, Profit Margin %). A Line chart of Total Sales by Order Date (month). A Bar chart of Total Profit by Sub-Category. A Donut or bar chart of Total Sales by Region. Add Slicers for Region and Category.
Format, interact, save	Add titles, apply one colour theme, align visuals to a grid. Click a bar to show cross-filtering across the page. Save as a .pbix file.

Teaching tip: Superstore contains sub-categories that lose money. Ask students to spot them in the bar chart. This gives them their first real insight, and it connects the tool to a business decision.

Part 3: Tableau Public hands-on

Goal: rebuild the same story in Tableau so students feel the difference between the two tools. Move faster here, because the ideas are already familiar.

Step	What to do
Connect and explore	Open Tableau Public > Connect to a text file and choose the CSV. Open a new sheet. Show that dimensions (blue) and measures (green) are listed on the left. Link this to the terms taught in Block 2.
Build three sheets by drag and drop	Sheet 1, trend: drag Order Date to Columns (set to Month) and Sales to Rows. Sheet 2, comparison: drag Sub-Category to Rows and Profit to Columns, sort descending, drag Profit to Color so losses stand out. Sheet 3, map: double-click State to create a map and drag Sales to Color.
Build the dashboard	Click New Dashboard. Drag the three sheets in. Use the "Use as Filter" icon on a chart so clicking it filters the others. Add a Region filter.
Save and compare	Show how to save to Tableau Public (a free account is required). Ask the class: which tool felt faster for exploring, and which felt easier for building measures?

Part 4: Dashboard design and storytelling

- **One dashboard, one question.** Do not put every chart on one page.
- **Put the most important numbers (KPIs) at the top left**, where people look first.
- **Choose the chart for the job:** line for trends over time, bar for comparing categories, a few slices only for parts of a whole.

- **Use colour with meaning**, for example red only for losses. Use at most 3 or 4 colours.
- **Give every chart a clear title**, label the axes, and remove clutter.

Storytelling in three parts: **Insight** (what the data shows), **Evidence** (the chart), **Action** (what the business should do). Example to try in class: find which sub-category loses money, show the chart, and suggest one action such as reviewing discounts on it.

Recap and quick check

Close with three take-aways: (1) clean data first, (2) a dashboard answers a question, (3) the skills transfer between Power BI and Tableau. Then ask these five questions, which students can answer by show of hands or in your quiz tool:

Question	Answer
1. What question does a data analyst answer?	What happened and why, from past data.
2. Name two common data problems.	Missing values, duplicates, wrong data types, inconsistent spelling.
3. Which Power BI tool is used to clean data?	Power Query (Transform Data).
4. What is the difference between a dimension and a measure?	A dimension is a category (Region); a measure is a number you add up (Sales).
5. Which chart shows a trend over time?	A line chart.